

meet unpredictable power conditions

3. It is user-friendly, with minimal and easy controls

4. If sensors like PIR are used, it must be EMI-/RFI-protected

5. Magnetic sensors are easy to align

6. The sound output is above 105 dB at 1 metre

Video/door phones. These can be installed at the entrance to see/talk to the person at the gate.

Cameras. A wide variety of cameras are available for home security. Your home can have a single camera at the front door or many cameras placed throughout the house. The multiple cameras can be attached to motion detectors and activated only when there is movement in a certain area. These can also be attached to VCRs or computers for additional monitoring and recording. If attached to a computer, you can check the status of the home from any-



Zicom Safe Homes home security solution

where over the Internet connection.

Cameras with audio are useful if you want to monitor sound as well as sight. These make use of a small microphone to monitor sound.

If the camera will be used outdoors, it should have a casing that is durable enough to withstand the weather. If you fear vandalism, an extra-durable casing is needed. For units that need to be plugged in, ensure that there is a power source nearby.

A CCTV will let you view the entrance and other strategic areas around your home. You may detect a regular unwanted or unknown face moving near your home.

Electronic/electromagnetic locks. Window and door lock systems can be added onto your home's existing locks,

creating a double lock at each point of entry. Intruders will find it difficult to break through such locks. More elaborate locks include fingerprint checks or number pads.

Panic switches. These are nothing but simple buttons with hooters attached to create an alarming sound of high pitch.

PIR lights. These lights activate automatically when there is an activity in the area of focus.

Standalone or integrated?

You can go for the different security devices one by one, depending on your budget. A gradual upgrade would also let you understand the use of each system one by one.

If your house has no inmate during the daytime, an integrated alarm system to prevent burglary, trespassing as well as fire would be required. For the elderly or kids alone at the home, a video surveillance system is preferable. For an occupied house visited by many outsiders, an audio/video doorphone is essential to identify the visitors before they get access inside the house.

If your house needs all the above, consider an integrated system. Generally, an audio doorphone coupled with a burglar alarm system with magnetic contacts at all openings, and panic switches at strategic locations, is the basic requirement for homes. Use of the rest of the equipment depends on individual requirements. For example, you can choose to add other sensors like PIR, glass break detectors and LPG detectors to the burglar alarm system.

Video doorphones are more popular. Builders generally provide them as a part of amenities in the society. You may also go for the technically advanced systems that provide clips of the visitors in your absence at home. There are also systems with smart card- and biometrics-based door ac-

cess. Emergency pushbutton lockets for the elderly in the house will alert the watchman of the society and the nearest hospital when they are in trouble.

Wired vs wireless

Both wired and wireless systems have their own advantages and disadvantages. The choice between the two has to be made based on the actual site conditions. For instance, homes that are under construction may go for a wired system, as the cables can be concealed. Moreover, wired systems are more cost-effective and more immune to any external disturbance. However, these require high degree of care and maintenance. Also, wiring is cumbersome.

On the other hand, wireless systems are versatile and flexible in operation. As the name suggests, these do not use hard cables for connectivity for various transmissions, but electric cable is still used for power source. You can go for them if you don't want to spoil the aesthetics and the décor of your well-furnished home with dangling cables.

But remember that wireless systems have limitations in terms of features set. No system with sophisticated features will be completely wireless. Even if it were, it would have issues related to signal strength, etc. Also, wireless systems based on advanced technology shall be extremely costly.

Wireless systems can be either RF- or WiFi-based. RF-based wireless systems are not very reliable in terms of signal strength and large data communication. They may also interfere with other electronic devices like mobile phone, television, radio and refrigerator in your home.

Therefore it is better to put up a wired system wherever possible.

Electric or battery powered?

No system will be entirely working on battery. It would need power to survive over longer period of time. Also, a system with fully featured